OTC-22 | UTILITIES & OPERATOR DEVELOPMENT

Water & Wastewater Plants: Mechanical O&M

SUGGESTED DURATION: 3 DAYS | IN-COMPANY / LIVE ONLINE

Improve mechanical operation, preventive maintenance and first-level troubleshooting of RO, DM, STP, ETP and ZLD utility equipment. The emphasis is safe routines and reliable equipment.

WHO SHOULD ATTEND

Water and wastewater operators, ITI/diploma technicians, mechanical maintenance staff and utility supervisors.

WHAT PARTICIPANTS WILL BE ABLE TO DO

- Identify major equipment and trace basic process flow and utility interfaces.
- Carry out structured operating observations and maintenance planning checks.
- Distinguish mechanical symptoms from issues requiring process-chemistry support.

SUGGESTED PROGRAMME OUTLINE

Day 1 | Plant and equipment foundations

Basic treatment flow and equipment roles; pumps, blowers, agitators, filters and valves; pressure, flow and level observations; lubrication, guards, leaks and housekeeping; operator rounds.

Day 2 | Maintenance and reliability

Pump and seal care; blower and bearing checks; filters and differential pressure; RO membrane system mechanical interfaces; PM routines, spares and isolation; equipment history and reporting.

Day 3 | Troubleshooting and teamwork

Low flow, high vibration, leakage and abnormal pressure; verify instruments and operating conditions; interface with process and laboratory teams; escalation and emergency awareness; practical operator action plans.

OTC-22 | PRACTICAL APPLICATION & DELIVERY

Water & Wastewater Plants: Mechanical O&M

PRACTICAL WORKSHOPS

Utility pump loses capacity

Use an illustrative log to separate suction restriction, air ingress, instrument error and downstream resistance.

Operator round redesign

Develop a short inspection route with normal observations, escalation triggers and clear shift handover.

PARTICIPANT TOOLS AND TAKEAWAYS

Operator round sheet; mechanical PM checklist; symptom/escalation guide; shift handover template.

HOW LEARNING IS DELIVERED AND ASSESSED

Facilitated explanations, worked examples, case discussions and small-group exercises. A short baseline check establishes starting knowledge; a closing knowledge check and practical review assess learning. Assessment can be adapted for operators, technicians or experienced engineers.

CUSTOMISATION AND CLIENT INPUTS

Plant flow diagram, equipment list and operator skill profile. Detailed chemical dosing, water chemistry, biological treatment and membrane process design require a process/chemistry specialist.

The agenda is indicative. Final content, duration, batch size, language and delivery arrangements are agreed after a training-needs discussion. Online delivery uses document reviews, calculations, role-play and case exercises; physical demonstrations require separate site arrangements.

FACILITATOR

Prem Prakash Mishra | Mechanical engineer, advisor and corporate trainer. Brings 39 years of career experience across Kuwait Oil Company, IPCL and NTPC, spanning projects, commissioning, maintenance, reliability, asset integrity and industrial audits. Specialist inputs, where needed, are agreed in the proposal.

COMPLETION AND FOLLOW-UP

An Omtech Corporation participation or completion certificate may be provided under the agreed assessment arrangements. This is company training, not an accredited professional qualification or a guarantee of certification. Post-training support is available with scope and duration agreed in the proposal.

REQUEST A CUSTOMISED TRAINING PROPOSAL

safehands@omtechcorporation.com

+91 9810772284 | +91 9987079111